INCH-POUND

MMM-A-181E
17 February 2026
SUPERSEDING
MMM-A-181D
January 23, 1980

FEDERAL SPECIFICATION

# ADHESIVES, PHENOL, RESORCINOL, OR MELAMINE BASE

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for the use of all Federal agencies.

## 1. SCOPE

1.1 Scope. This specification covers general-purpose, 2-part adhesives, or optionally a 1-part adhesive for type III (see 3.1.1), for wood assembly gluing. The adhesives are for the manufacture of laminated members and other wooden articles where a high-strength, durable adhesive is required.

1.2 Classification.

1.2.1 Types, grades, and classes. The adhesive covered by this specification shall be of the following types, grades, and classes (see 6.2):

1.2.1.1 Type I. Room-temperature setting, 24° to 35°C (75° to 95°F), glue-line temperature.

Grade A - Two years storage life.

Grade B - Six months storage life.

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army DEVCOM, Army Research Laboratory, Army Research Directorate, Specifications and Standards Office, Attn: FCDD-RLA-M, Aberdeen Proving Ground, MD 21005-5069. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A FSC 8040

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1.2.1.2 Type II. Intermediate-temperature setting, 35° - 88°C (95° - 190°F), glue-line temperature.

Grade C - Six-months storage life.

Class 1 - Liquid.
Class 2 - Powder.

1.2.1.3 Type III. High-temperature setting, 88 - 149°C (190° to 300°F), glue-line temperature.

Grade E - One year storage life.

Class 1 - Liquid.
Class 2 - Powder.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

DEPARTMENT OF DEFENSE STANDARDS

| | | |
|---|---|---|
| MIL-STD-129 | - | Marking for Shipment and Storage |
| MIL-STD-1916 | - | DoD Preferred Methods for Acceptance of Product |

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

| | | |
|---|---|---|
| FED-STD-313 | - | Material Safety Data, Transportation Data, and Disposal Data for Hazardous Materials Furnished to Government Activities. |

(Copies of this document are available online at https://quicksearch.dla.mil/).

DEPARTMENT OF TRANSPORTATION (DOT)

| | | |
|---|---|---|
| 49 CFR 100-199 | - | Rules and Regulations for the Transportation of Hazardous Materials |

(Copies of this document are available online at https://www.ecfr.gov/current/title-49).



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2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

THE AMERICAN NATIONAL STANDARDS INSTITUTE

| | | |
|---|---|---|
| ANSI/ASQ Z1.4 | - | Sampling Procedures and Tables for Inspection by Attributes |

(Copies of these documents are available online at https://ansi.org/).

ASTM INTERNATIONAL

| | | |
|---|---|---|
| ASTM D905 | - | Standard Test Method for Strength Properties of Adhesive Bonds in Shear by Compression Loading |
| ASTM D906 | - | Standard Test Method for Strength Properties of Adhesives in Plywood Type Construction in Shear by Tension Loading |
| ASTM D1151 | - | Standard Practice for Effect of Moisture and Temperature on Adhesive Bonds |
| ASTM D1583 | - | Standard Test Method for Hydrogen Ion Concentration of Dry Adhesive Films |

(Copies of these documents are available online at https://www.astm.org/).

2.4 Order of precedence. Unless otherwise noted herein or within the contract or purchase order, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 Material. The material used in the manufacture of the adhesive shall be such as to produce finished adhesive conforming to this specification.

3.1.1 Condition. The adhesive shall be furnished as a liquid with separate hardener, or when specified (see 6.2), as a powder with separate hardener. The hardener for the type III adhesive may be incorporated as a component part of the resin.

3.2 Resins. Only resins based on phenol, melamine, or resorcinol shall be used in manufacturing the adhesive.

3.3 Insoluble matter. The quantity of insoluble matter, including filler and any insoluble partially polymerized resin, shall be not more than 20 percent by weight of the nonvolatile constituents of the adhesive when mixed and ready for use (see 4.5.1).



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3.4 Hardener. Hardener shall be supplied in either powder or liquid form. Where the hardener is furnished in powder form, it may be combined with part or all of the filler. The hardener shall be mixed with the adhesive prior to application to the joint.

3.5 Filler. Where filler is used it shall be an inert, insoluble powder incapable of being dissolved in or being swelled excessively by water. The filler may be combined with the resin or supplied separately (see 3.3). Amylaceous fillers such as flour or starch, and protein fillers shall not be permitted (see 4.5.2).

3.6 pH of set film. The set adhesive film shall develop a pH value of not less than 3.5 nor more than 10.5 when determined as specified in 4.5.3.

3.7 Cleanability. The mixed adhesive shall be capable of being cleaned from spreading equipment and containers when tested as specified in 4.5.4.

3.8 Liquid working life. Adhesive shall have a minimum working life of 2.5 hours at 23.9° ± 2.8°C (75° ± 5°F), when determined as specified in 4.5.5.

3.9 Curing cycle. The curing cycle of the mixed adhesives to cure to the hardened state for the tests specified in 4.5.6 and 4.5.7 shall be as follows:

Type I - The curing cycle of type I adhesive shall be in accordance with 4.5.6.1.1 or 4.5.7.1.1, as applicable.

Type II - The curing cycle of type II adhesive shall be not longer than 5 hours and with a temperature not higher than 88°C (190°F) measured in the adhesive line.

Type III - The curing cycle of type III adhesive shall be not longer than 5 hours and with a temperature not higher than 149°C (300°F) measured in the adhesive line.

3.10 Shear strength by compression loading (type I and II only). The types I and II adhesive shall develop an average-shear strength of not less than 19.31 MPa (2,800 psi), using hard maple blocks when tested as specified in 4.5.6.

3.11 Plywood shear strength.

3.11.1 Dry. The adhesive shall develop an average strength of not less than 2.76 MPa (400 psi) when tested as specified in 4.5.7.2.

3.11.1 Wet. The adhesive shall develop an average strength of not less than 2.76 MPa (400 psi) when tested as specified in 4.5.7.3.

3.12 Instructions. A label or tag with the following instructions shall be attached to each container or adhesive:

(a) Brief instructions for the storage and care of the adhesive prior to use.

(b) The date of manufacture (by month and year, not by code) and the maximum storage life recommended by the manufacturer.



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(c) Instructions for mixing of the adhesive and working life of the mixed adhesive, including the manufacturer's recommendations on the maximum and minimum assembly period, for both open and closed assembly, for at least 3 temperatures, covering typical operating conditions such as 23, 24 and 29°C (70, 75 and 85°F).

(d) The proper use of the adhesive (see 3.9), detailing the limitations within which recommended pressure periods may be varied by the use of elevated temperatures during pressing and curing periods.

(e) In addition, each tag or label shall contain the following:

"Due to the sensitivity of the adhesive to extremes of temperature, it is recommended that the adhesive be kept out of the sun. All adhesive held in stock should be stored dry, tightly covered, and where practicable, at temperatures below 27°C (80°F)."

"Avoid inhaling vapors generated while mixing 2-part adhesive."

"SKIN AND EYE IRRITANT - This material is irritating to the skin. Avoid unnecessary skin contact. Wash frequently. Do not allow material to harden on skin. If material enters eyes, flush immediately with large quantities of water."

3.13 Flammable liquid. When applicable, material classified as a flammable liquid in accordance with the Department of Transportation document 49 CFR 100-199 (Code of Federal Regulations) shall also be labeled in accordance with the appropriate regulations.

3.14 Storage life. The adhesive shall be suitable for use and shall meet the requirements in 3.8, 3.9, 3.10 and 3.11, after having been stored in accordance with 4.5.8 for the respective period as follows:

| | | |
|---|---|---|
| Grade A | - | 2 years from date of manufacture. |
| Grade B | - | 6 months from date of manufacture. |
| Grade C | - | 6 months from date of manufacture. |
| Grade E | - | 12 months from date of manufacture. |

3.15 Certification. When specified, the contractor shall submit to the procuring agency a letter stating that the resins comply with 3.3 and that storage life shall conform to 3.14. The letter shall state whether the contractor has performed the storage life test (see 4.5.8). The letter shall be signed by a responsible agent of the certifying organization aid shall be accompanied evidence of this agent's authority to bind his principal.

3.16 SDS. The manufacturer shall comply with the requirements as specified in the OSHA Brief:

(http://www.osha.gov/dsg/hazcom/osha-brief.html

and as specified in Appendix D of 29 CFR 1910.1200 see:

(https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.1200AppD).



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A SDS shall be prepared for the material in accordance with FED-STD-313 and forwarded to the qualifying activity (see 6.4.3). The SDS shall be included with each shipment of the material covered by this specification and submitted to pertinent Government agencies as stated in FED-STD-313.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein.

4.2 Lot. Unless otherwise specified by the procuring agency, for purposes of sampling, a lot shall consist of adhesive of one component, type, grade and class, manufactured at one time, as one batch and offered for delivery at one time. When material cannot be identified by batch, a lot shall consist of not more than 1,125 kg (2,500 pounds) of adhesive of one component, type, grade, and class offered for delivery at one time.

4.3 Sampling.

4.3.1 For examination. A random sample of filled containers shall be selected for examination in accordance with MIL-STD-1916 and surveyed in accordance with level I of ANSI/ASQ Z1.4.

4.3.2 For tests. A representative 0.95 liter (1 quart) sample of adhesive and a sufficient quantity of hardener shall be taken from each lot. All of the tests shall be performed on this sample.

4.4 Examination of packaging. An examination shall be made to determine compliance with the requirements of Section 5. The sample unit shall be one shipping container fully prepared for delivery. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 expressed in terms of percent defective.

4.5 Tests.

4.5.1 Nonvolatile and insoluble content.

4.5.1.1 Nonvolatile content. Mix at least 100 g of adhesive according to the manufacturer's directions, weighing to the nearest 0.01 g. Place approximately 10g of fine, oven-dried, quartz sand in a Petri dish together with a small glass stirring rod, and weigh the dish and its contents to the nearest milligram (mg) on an analytical balance; add approximately 2g of the mixed adhesive, and weigh again to the nearest mg. Mix the adhesive and sand intimately by means of the stirring rod and distribute uniformly over the bottom of the dish in as thin a layer as possible. Heat the dish, rod and contents overnight in an oven at 100°+ 5°C (212°+ 90°F), cool, reweigh, and calculate the percent nonvolatile as follows:

$$\text{Percent nonvolatile} = \frac{\text{(weight of adhesive loss in weight)}}{\text{weight of adhesive}} \times 100$$

4.5.1.2 Insoluble matter. Make this determination on the same batch of mixed adhesive used for the determination of nonvolatile content (see 4.5.1.1).



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4.5.1.2.1 Phenol and resorcinol resins. Weigh approximately 2 g of the freshly mixed adhesive to the nearest mg in a 100 milliliter (ml) beaker: add 50 ml of absolute methyl alcohol; stir for 3 minutes; then filter through a previously weighed, sintered-glass crucible of medium porosity. Transfer all residue to the filter with a second 50 ml portion of methyl alcohol; wash the crucible and residue with approximately 30 ml of additional alcohol; dry for 2 hours in an oven at 100° ± 5°C (212° ± 9°F); cool in a desiccator; and weigh to the nearest mg. Calculate the insoluble content as a percent of the nonvolatile content as follows:

$$\text{Percent insoluble} = \frac{\text{weight of residue} \times 100 \times 100}{\text{weight of adhesive} \times \text{percent nonvolatile}}$$

4.5.1.2.2 Melamines. Weigh approximately 2 g of the freshly mixed adhesive to the nearest mg in a 100-ml beaker; add 50 ml of a solvent composed of 20 parts of glacial acetic acid, 20 parts of 95-percent ethyl alcohol, and 60 parts of distilled water by volume. Stir the adhesive and solvent for 1 minute, then add 0.5 g of diatomaceous earth (previously acid-washed, oven dried at 100°C ± 5°C (212°+ 9°F) and stored in a desiccator). Continue the stirring for 2 minutes longer, then filter through a previously weighed, sintered-glass, crucible of medium porosity. Wash all residue remaining in the beaker onto the filter with at least 30 ml of the solvent, then wash with cold water to remove the acid and finally wash with 20 ml of 95-percent ethyl alcohol. After preliminary drying by suction, dry the crucible with the residue in an oven at 100°C ± 5°C (212°+ 9°F) for 2 hours, cool in a desiccator, and weigh to the nearest mg. Calculate the insoluble content as percent of nonvolatile content as follows:

$$\text{Percent insoluble} = \frac{(\text{weight of residue} - 0.5) \times 100 \times 100}{\text{weight of adhesive} \times \text{percent nonvolatile}}$$

4.5.2 Amylaceous matter.

4.5.2.1 Phenol and resorcinal resins. To approximately 0.5 g of mixed adhesive add approximately 5 ml of 95-percent ethyl alcohol in a test tube of small diameter, shake thoroughly, allow insoluble matter to settle, and decant the supernatant liquid. Repeat this process twice with alcohol and three times with distilled water. To the residue after the final decantation, add a drop of a solution made of 5 g of iodine, 5 g of potassium iodide, and 100 ml of distilled water. A pronounced darkening of the residue indicates the presence of amylaceous matter.

4.5.2.2 Melamine resins. Use the same procedure as specified in 4.5.2.1 except that, instead of the ethyl alcohol, use a solution made up of 20-parts glacial acetic acid, 20-parts ethyl alcohol, and 60-parts distilled water by volume. When the melamine adhesive is supplied with separate hardener, ethyl alcohol shall be used for the unmixed hardener.

4.5.3 pH of set film. The pH of the set film shall be determined as specified in ASTM D1583.

4.5.4 Cleanability. A portion of the mixed adhesive shall be placed in a glue pot, and before the expiration of its working life and at the proper temperature of use as furnished by the manufacturer, shall be completely cleaned from the pot by water, 0.5 percent sodium hydroxide solution, alcohol, or a mixture of these.



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4.5.5 Liquid-working life. Pour approximately 200 g of freshly mixed adhesive into a large glass test tube or beaker approximately 35 millimeters in diameter, provided with a glass stirring rod, and immerse the tube with contents in a water bath held at 23.9° ± 2.8°C (75° ± 5°F). Stir the adhesive frequently with the rod and note the time at which the adhesive is confirmed as no longer spreadable on wood surface with a stiff bristle brush. The time to reach this consistency after mixing is the liquid working life of the adhesive at 23.9° ± 2.8°C (75° ± 5°F).

4.5.6 Shear strength by compression loading (type I and II only).

4.5.6.1 Test specimens. At least 4, separated, hard maple wooden test blocks shall be prepared in accordance with ASTM D905 except that at least 2 of the test blocks shall be prepared for the minimum assembly time and 2 for the maximum assembly time recommended by the manufacturer, after which 1.73 MPa (250 psi) pressure shall be applied.

4.5.6.1.1 Type I. The blocks shall then be held under this pressure for 7 hours at a temperature of 23.9° ± 2.8°C (75° ± 5°F). After releasing the pressure, condition the test blocks in accordance with ASTM D905.

4.5.6.1.2 Type II. The blocks shall be cured according to the instructions of the manufacturer and in accordance with 3.9. After being cured, the blocks shall be allowed to cool to room temperature before the pressure is released. After releasing the pressure, condition the test blocks in accordance with ASTM D905.

4.5.6.2 Procedure. After conditioning, the test blocks shall be cut and tested in accordance with ASTM D905. Whenever 50 percent or more specimens fail at a load less than 19.3 MPa (2,800 psi) and failure occurs in the wood fiber, the specimen shall be disregarded in computing the average shear strength. If 5 or more specimens fail in this category, a new set of test blocks and specimens shall be prepared and tested. If the average shear strength is less than 19.3 MPa (2,800 psi) the lot shall be rejected.

4.5.7 Plywood shear strength.

4.5.7.1 Test specimens. At least 9 separate 3-ply plywood test panels shall be prepared in accordance with ASTM D906 except that 3 of the 3-ply panels shall be left in closed assembly for the minimum assembly time and 6 for the maximum assembly time recommended by the manufacturer, after which they shall be placed under pressure of 1.03 to 1.38 MPa (150 to 200 psi).

4.5.7.1.1 Type I. The panels shall be allowed to remain under pressure for at least 7 hours at a temperature of 23.9° ± 2.8°C (75° ± 5°F). After releasing the pressure, condition the test-panels, cut-into test specimens, and number according to ASTM D906.

4.5.7.1.2 Type II and III. The panels shall be cured according to the instructions of the manufacturer and in accordance with 3.9. The panels shall then be removed from the press and cooled to room temperature. Panels shall then be conditioned, cut into test specimens, and numbered in accordance with ASTM D906.



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4.5.7.2 Dry shear. The specimens number 1, 4 and 7 from each panel shall be tested dry in accordance with ASTM D906.

4.5.7.3 Wet shear.

4.5.7.3.1 Boiling water. The specimens numbered 2, 5 and 8 shall be immersed in boiling water for 3 hours ± 10 minutes, removed, and immersed in water at 23.9° ± 2.8°C (75° ± 5°F) for 30 ± 5 minutes and then tested immediately upon removal from the water. Test specimens from the same sample may be immersed in one container but specimens from different samples shall be immersed in separate containers. The specimens shall then be tested in accordance with ASTM D906.

4.5.7.3.2 Cold water. The specimens numbered 3, 6 and 9 shall be subjected to an immersion in water at 23.9° ± 2.8°C (75° ± 5°F) for 48 ± 2 hours as specified in ASTM D1151 and immediately tested for shear strength in accordance with ASTM D906.

4.5.7.4 Test reports. Plywood shear-test reports shall include average load at failure, average percent of wood failure, and the number of specimens from which the average was computed. Whenever a specimen fails at a load less than that specified (see 3.10), the failure taking place in 50 percent or more in the wood fiber, the specimen shall be disregarded in computing the average values required.

4.5.8 Storage life. Specimens of adhesive shall be stored in a tightly sealed container for the period required in 3.14 at 23.9° ± 2.8°C (75° ± 5°F) and then tested for the requirements as specified in 3.14.

## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2). When packaging of materiel components is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Markings. Containers of adhesive shall include labels that identifiy the specification number, type, grade and class so that traceability of the lot can be verified and a SDS (see 3.16) shall be included.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)



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6.1 Intended use. The adhesives are used in the manufacture of laminated members and other wooden articles where a high-strength, durable adhesive is required.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Whether powder form is required (see 1.2.1).
c. Type and grade (see 1.2.1).
d. Size of container required (see 5.1).
e. Selection of the applicable levels of packaging and packing (see 5.1).
f. Special labelling and marking if required (see 3.12, 5.2, and 6.4).
g. Preparation of SDS in accordance with FED-STD-313 for the adhesive and inclusion of SDS with shipment of material (see 3.16).

6.3 Health and safety considerations. Users are cautioned that because of the formaldehyde content of the adhesive the usual cleanliness standards should be employed in its use. When users have experienced adhesive contact with the skin, it has been found that the use of protective creams and frequent washing alleviates tendency toward industrial dermatitis unless, of course, the particular user is allergic or sensitized to formaldehyde. It should be emphasized that many industrial dermatitis problems are solved by proper onsite cleanliness on the part of the worker, conscientious use of personal protective equipment (PPE) and adequate worksite ventilation. Meticulous workspace housekeeping and constant employee awareness should be maintained to guard against the two major dangers in resin operations:

(1) Skin contact with the resins or other materials.
(2) Exposure to their vapors.

The following rules should be adhered to:

(1) Inform workers of possible hazards.
(2) Provide ventilation to control vapors produced when mixing the resins and hardener as well as to control the particulates during tooling, grinding, or shaping.
(3) Maintain good personal hygiene procedures, provide and enforce the use of appropriate personal protective clothing and equipment (i.e. gloves, safety eyewear, eye lavage and safety showers).
(4) For additional safety recommendations refer to the SDS (see 3.16 and 6.4).
(5) Keep containers closed when not in active use.

6.4 SDS. Contracting officers will identify those activities requiring copies of a completed SDS, when implemented, prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.



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6.5 Subject term (key word) listing.

| | | |
|---|---|---|
| Adhesion | Cure | Glue |
| Bonding | Fibers | Resins |

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.



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## CONCLUDING MATERIAL

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Navy - YD

Preparing activity:
ARMY - MR

(Project 8040-2025-005)

Review activities:
Army - CR, GL, MD, MI
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